

Lecture 9: Bay's Theorem

29 June 2026 12:40

Topics of this lecture:

1. Bay's Theorem

According to conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

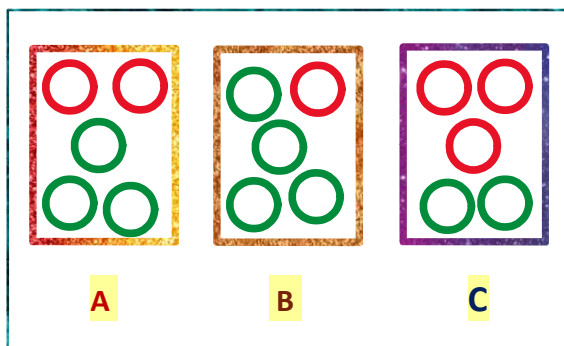
According to multiplication theorem:

$$P(A \cap B) = P(B | A) \cdot P(A)$$

Now we can write:

$$P(A|B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

This is the expression of Bay's Theorem



Q. Pick a bag at random
And draw a random ball out of it.

1. Probability of getting a red ball given bag a is chosen ?

Conditional Probability

$$P(\text{Red Ball} | A) = 2/5$$

2. Probability of getting a red ball and choosing bag A ?

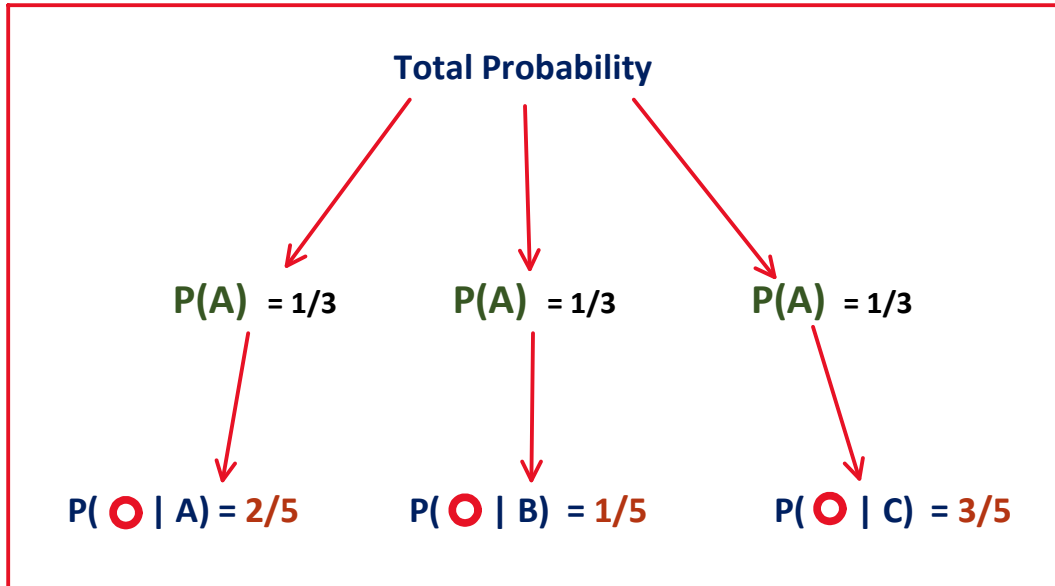
Multiplication Theorem

$$\begin{aligned} P(\text{Red Ball} \cap A) &= P(\text{Red Ball} | A) \cdot P(A) \\ &= 2/5 \cdot 1/3 \\ &= 2/15 \end{aligned}$$

3. Probability of getting a red ball ?

Total Probability

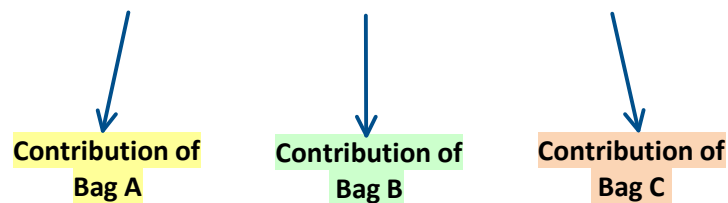
$$\begin{aligned}
 P(\text{Red}) &= P(\text{Red} | A) \cdot P(A) + P(\text{Red} | B) \cdot P(B) + P(\text{Red} | C) \cdot P(C) \\
 &= \frac{2}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} + \frac{3}{5} \cdot \frac{1}{3} \\
 &= \frac{2}{5}
 \end{aligned}$$



4. Probability of bag a is chosen given that the drawn ball is red ?

Total Probability

$$P(\text{Red}) = P(\text{Red} | A) \cdot P(A) + P(\text{Red} | B) \cdot P(B) + P(\text{Red} | C) \cdot P(C)$$



$$P(A | \text{Red}) = \frac{P(\text{Red} \cap A)}{P(\text{Red})} = \frac{\text{Contribution of Bag A}}{\text{Total Probability of red ball}}$$

$$\begin{aligned}
 P(A | \text{Red}) &= \frac{\frac{2}{5} \cdot \frac{1}{3}}{\frac{2}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} + \frac{3}{5} \cdot \frac{1}{3}} \\
 &= \frac{2/15}{2/5} = \frac{2}{5}
 \end{aligned}$$

$$= \frac{2}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} + \frac{3}{5} \cdot \frac{1}{3}$$

$$P(A \mid \text{Red}) = \frac{\frac{2}{15}}{\frac{2}{5}} = \frac{1}{3}$$